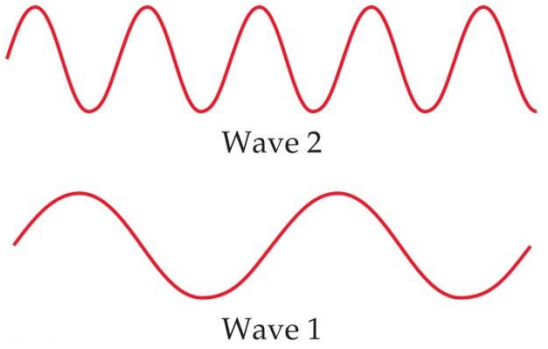
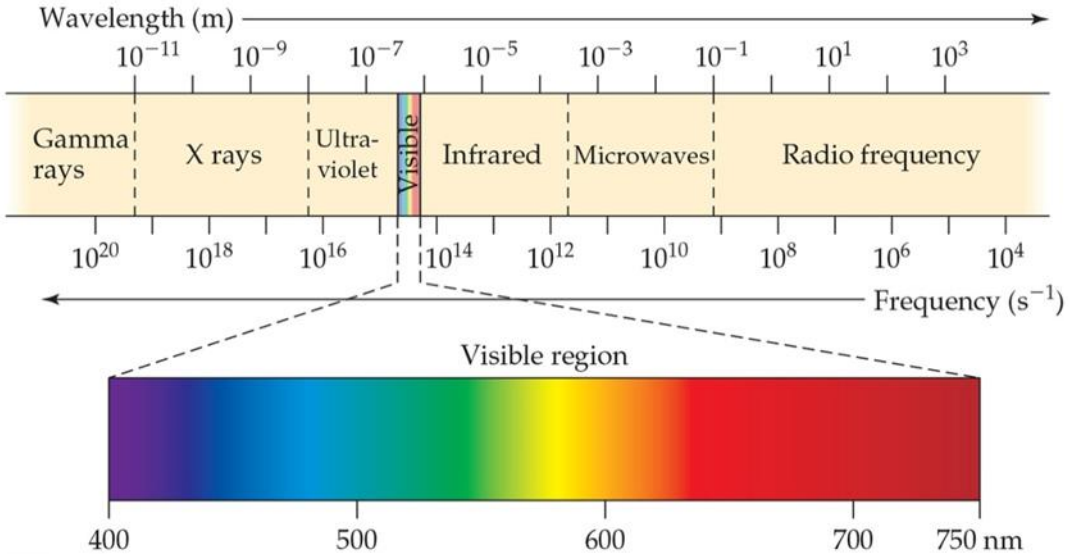
EXERCISES ABOUT CHAPTER 6, 8

- Two electromagnetic waves are represented in the margin. **(a)** Which wave has the higher frequency? **(b)** If one wave represents visible light and the other represents infrared radiation, which wave is which?





Wave 1 has a longer wavelength (greater distance between peaks). The longer the wavelength, the lower the frequency ($\nu = c/\lambda$).

Thus, Wave 1 has the lower frequency, and Wave 2 has the higher frequency.

The electromagnetic spectrum indicates that infrared radiation has a longer wavelength than visible light. Thus, Wave 1 is infrared radiation.

- The yellow light given off by a sodium vapor lamp used for public lighting has a wavelength of 589 nm. What is the frequency of this radiation?

$$\nu = \frac{c}{\lambda} = \left(\frac{3.00 \times 10^8 \,\text{m/s}}{589 \,\text{nm}}\right)\left(\frac{1 \,\text{nm}}{10^{-9} \,\text{m}}\right)$$

$$= 5.09 \times 10^{14} \,\text{s}^{-1}$$

When we insert the values for *c* and λ, we note that the units of length in these two quantities are different. We use a conversion factor to convert the wavelength from nanometers to meters.

- Calculate the energy of one photon of yellow light that has a wavelength of 589 nm.

$$\nu = (3.00 \times 10^{8}\,\text{m/s})/(589 \times 10^{-9}\,\text{m}) = 5.09 \times 10^{14}\,\text{s}^{-1}$$

$$E = (6.626 \times 10^{-34}\,\text{J-s})(5.09 \times 10^{14}\,\text{s}^{-1}) = 3.37 \times 10^{-19}\,\text{J}$$

If one photon of radiant energy supplies 3.37×10^{-19} J, we calculate that one mole of these photons will supply:

$$(6.02 \times 10^{23}\,\text{photons/mol})(3.37 \times 10^{-19}\,\text{J/photon})$$
$$= 2.03 \times 10^{5}\,\text{J/mol}$$

- Which of the following expressions correctly gives the energy of a mole of photons with wavelength λ?

(a) $E = \frac{h}{\lambda}$ **(b)** $E = N_A\frac{\lambda}{h}$ **(c)** $E = \frac{hc}{\lambda}$ **(d)** $E = N_A\frac{h}{\lambda}$ **(e)** $E = N_A\frac{hc}{\lambda}$

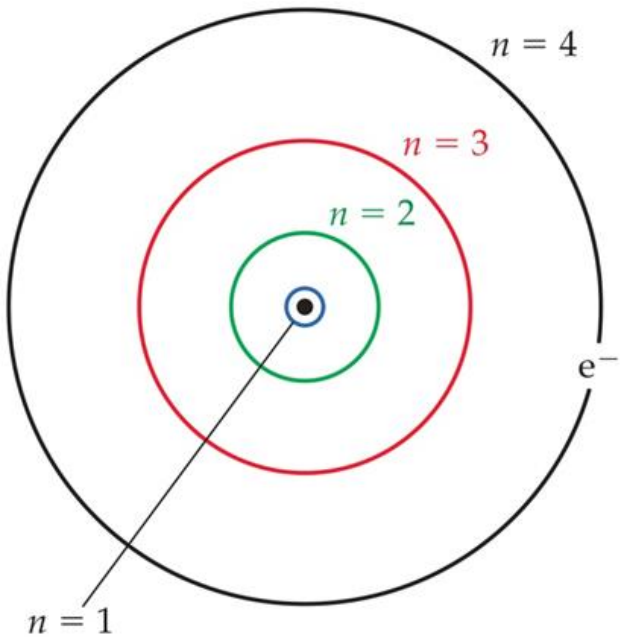
- Energy with 215 kj/mol of photon, calculate the wavelength as nm.

$$E = h \cdot v = \frac{h \cdot c}{\lambda}$$

$$\lambda = \frac{h \cdot c}{E}$$

$$\lambda = \frac{6.626 \times 10^{-34} j \cdot s \times 3 \times 10^{8}\, m/s}{215 \frac{kj}{mol} \times \left(\frac{1000 j}{1 kj}\right) \times \left(\frac{1 mol}{6.022 \times 10^{23} photon}\right)} = 5.57 \times 10^{-7} m \times \left(\frac{10^{-9} nm}{1 m}\right) = 557\, nm$$

- In the Bohr model of the hydrogen atom, electrons are confined to orbits with fixed radii, and those radii can be calculated. The radii of the first four orbits are 0.53, 2.12, 4.76, and 8.46 Å, respectively, as depicted below.
  (a) If an electron makes a transition from the $n_i = 4$ level to a lower-energy level, $n_f = 3$, 2, or 1, which transition would produce a photon with the shortest wavelength?
  (b) What are the energy and wavelength of such a photon, and in which region of the electromagnetic spectrum does it lie?



(a) The wavelength of a photon is related to its energy through the relationship $E = h\nu = hc/\lambda$.
Hence, the photon with the smallest wavelength will have the largest energy.

The electron loses the most energy on transitioning from the $n_i = 4$ state to the $n_f = 1$ state, and the photon emitted in that transition has the highest energy and the smallest wavelength.

(b) $$\Delta E = -2.18 \times 10^{-18}\,\text{J}\left(\frac{1}{1^2} - \frac{1}{4^2}\right) = -2.04 \times 10^{-18}\,\text{J}$$

$$E_{\text{photon}} = -\Delta E = 2.04 \times 10^{-18}\,\text{J}$$

Next we rearrange Planck's relationship to calculate the frequency of the emitted photon.

$$\nu = E/h = (2.04 \times 10^{-18}\,\text{J})/(6.626 \times 10^{-34}\,\text{J-s}) = 3.02 \times 10^{15}\,\text{s}^{-1}$$

Finally, we use the frequency to determine the wavelength.

Light with this wavelength falls in the ultraviolet region of the electromagnetic spectrum.

$$\lambda = c/\nu = (2.998 \times 10^{8}\,\text{m/s})/(3.02 \times 10^{15}\,\text{s}^{-1})$$
$$= 9.72 \times 10^{-8}\,\text{m} = 97.2\,\text{nm}$$

- Bohr hydrogen atom model if one electron transfers from n=5 to n=3, calculate ΔE and frequency of spectral line.

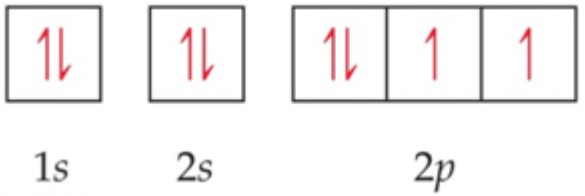
$$\Delta E = -2.18 \times 10^{-18} j \left(\frac{1}{{n_f}^2} - \frac{1}{{n_i}^2} \right)$$

$$\Delta E = -2.18 \times 10^{-18} j \left(\frac{1}{3^2} - \frac{1}{5^2} \right) = -1.55 \times 10^{-19} j$$

$$v = \frac{E}{h} = \frac{1.55 \times 10^{-19} j}{6.626 \times 10^{-34} j \cdot s} = 2.34 \times 10^{14} s^{-1}$$

- Draw the orbital diagram for the electron configuration of oxygen, atomic number 8. How many unpaired electrons does an oxygen atom possess?

Two electrons each go into the 1*s* and 2*s* orbitals with their spins paired. This leaves four electrons for the three degenerate 2*p* orbitals.
Following Hund's rule, we put one electron into each 2*p* orbital until all three orbitals have one electron each. The fourth electron is then paired up with one of the three electrons already in a 2*p* orbital,

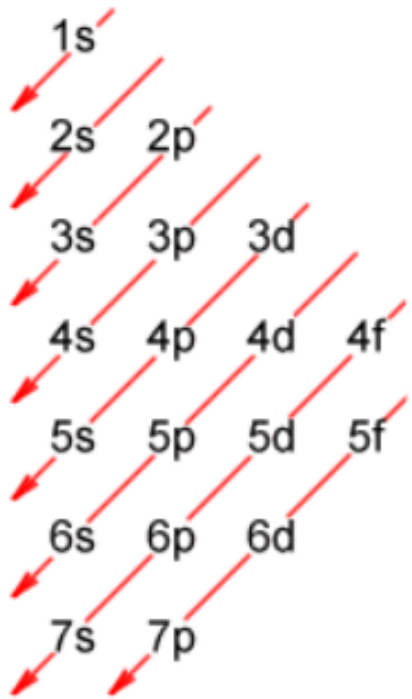


The corresponding electron configuration is written $1s^2 2s^2 2p^4$.
The atom has two unpaired electrons.

- Write the electron configuration for silicon, element 14, in its ground state.

$[Ne]\ 3s^2\ 3p^2$





1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, 3*d*, 4*p*, 5*s*, 4*d*, 5*p*, 6*s*, 4*f*, 5*d*, 6*p*, 7*s*, 5*f*, 6*d*, 7*p*

- Predict the ion generally formed by **(a)** Sr, **(b)** S, **(c)** Al.

(a) Strontium is a metal in group 2 and therefore forms a cation. Its electron configuration is [Kr]$5s^2$, and so we expect that the two valence electrons will be lost to give an Sr^{2+} ion.

(b) Sulfur is a nonmetal in group 16 and will thus tend to be found as an anion. Its electron configuration ([Ne]$3s^2 3p^4$) is two electrons short of a noble-gas configuration. Thus, we expect that sulfur will form S^{2-} ions.

(c) Aluminum is a metal in group 13. We therefore expect it to form Al^{3+} ions.

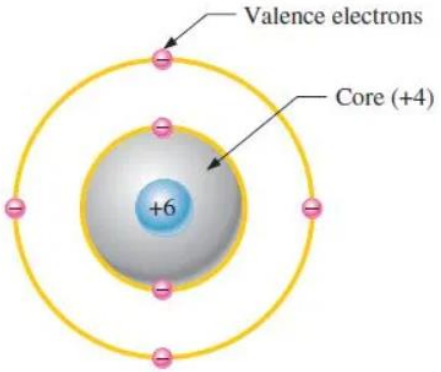
- Which of these elements is most likely to form ions with a 2+ charge?

(a) Li **(b)** Ca **(c)** O **(d)** P **(e)** Cl

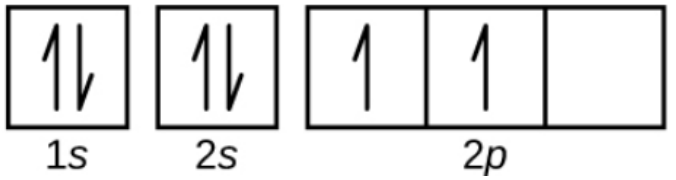


- Give Lewis structure for C_2N_2

1.Find the sum of valence electrons of all atoms in the polyatomic ion or molecule.



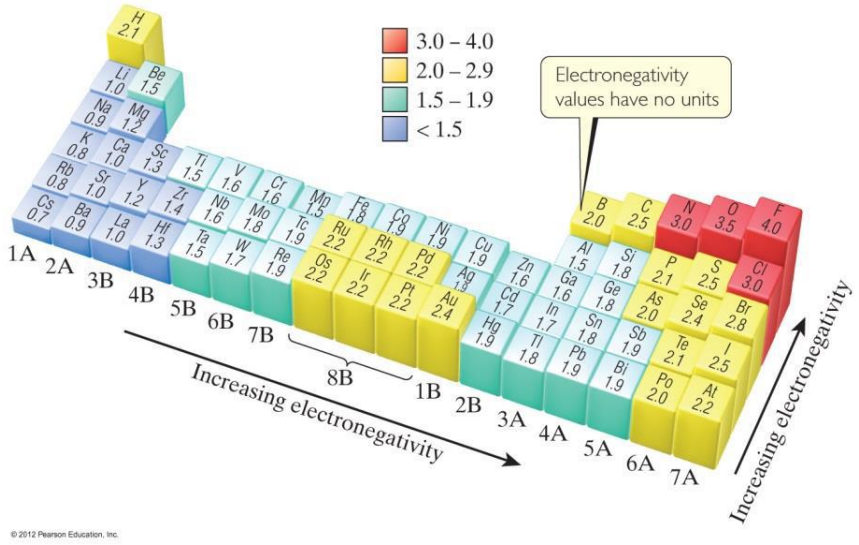
C $1s^2 2s^2 2p^2$



N $1s^2 2s^2 2p^3$

Sum of valence electrons $(2 \times 4) + (2 \times 5) = 18e^-$

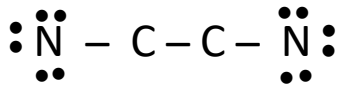
2.The central atom is the *least* electronegative element that isn't hydrogen. Connect the outer atoms to it by single bonds.



$$N - C - C - N$$

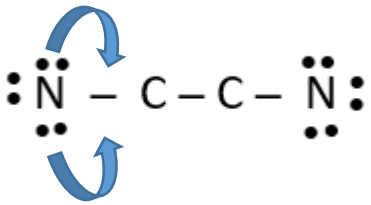
$$18e^- - 6e^- = 12e^-$$

3.Fill the octets of the outer atoms.



$$12e^- - 12e^- = 0$$

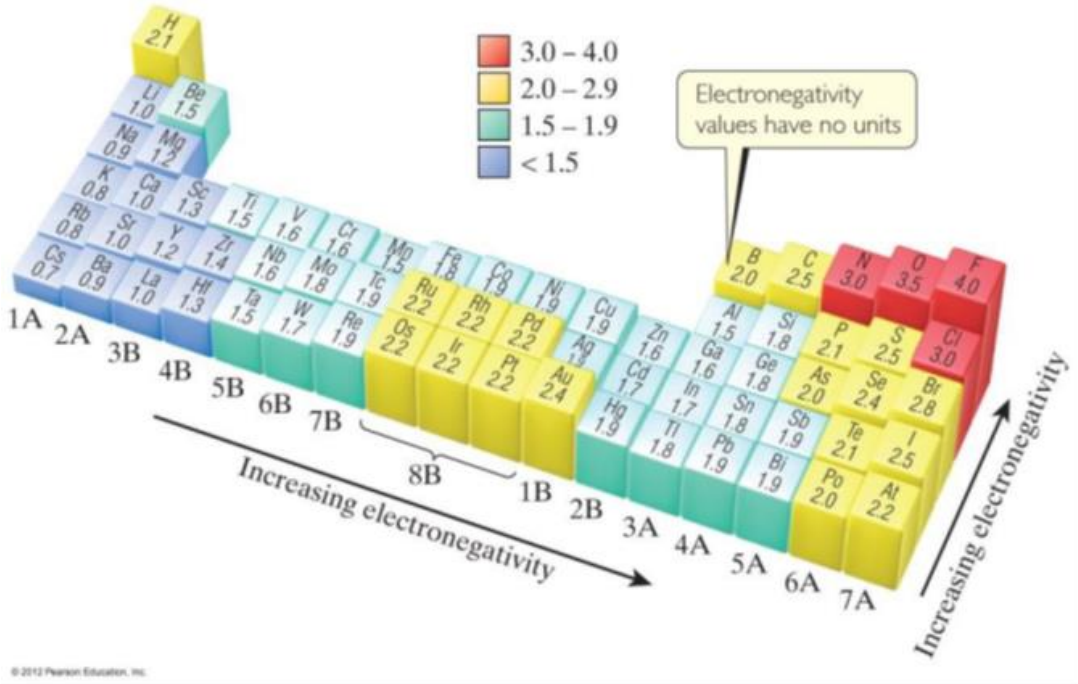
5. If you run out of electrons before the central atom has an octet…
…form multiple bonds until it does.



$$N \equiv C - C \equiv N$$

- Which bond is more polar? B—Cl or C—Cl

The chlorine atom is common to both bonds.
Therefore, we just need to compare the electronegativities of B and C.



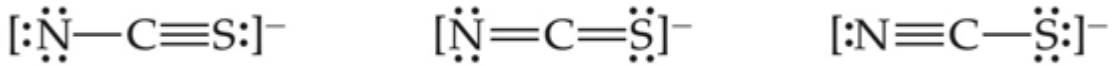
Boron is to the left of Carbon in the periodic table

Boron has the lower electronegativity

More polar bond will be the one between the atoms with the biggest differences in electronegativity.

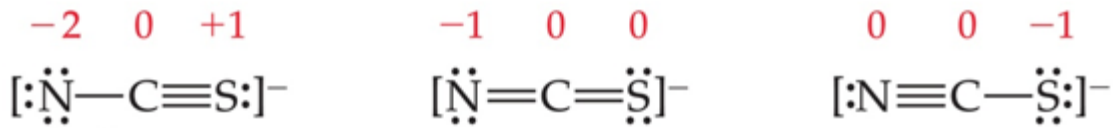
Consequently, the B—Cl bond is more polar

- Three possible Lewis structures for the thiocyanate ion, NCS^-, are



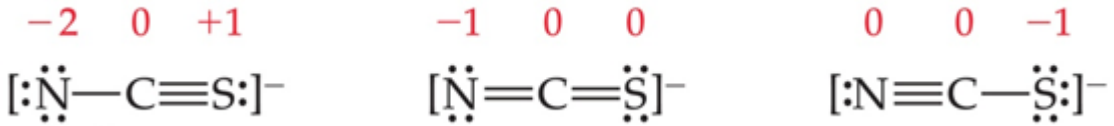
(a) Determine the formal charges in each structure.
(b) Based on the formal charges, which Lewis structure is the dominant one?

(a) We can determine the formal charges in the three structures



How to identify the Dominant Lewis structure

1. The dominant Lewis structure is generally the one in which the atoms bear formal charges closest to zero.
2. A Lewis structure in which any negative charges reside on the more electronegative atom is generally more dominant than one that has negative charges on the less electronegative atoms.



(b) The dominant Lewis structure generally produces formal charges of the smallest magnitude (guideline 1).

That eliminates the left structure.

N is more electronegative than C or S.
Therefore, we expect any negative formal charge to reside on the N atom (guideline 2).

For these two reasons, **the middle Lewis structure** is the dominant one for NCS^-.